A Structured Questioner for Collaborative Instance Navigation: Prompt Design, Data Construction, and Model Selection for the CoIN Challenge 2026

Ning Yang

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Abstract

We present a questioner for the CoIN Challenge 2026. The agent is Qwen3-VL-32B-Instruct, fine-tuned with two-stage QLoRA on (i) synthetic instance pairs labeled under a frozen structured prompt and (ii) ground-truth-validated pairs from the official training episodes. A category-only question-deduplication rule is applied at inference. Development follows a pre-registered ranking of full-success rate (FR), then success rate (SR), then questions per observation (NQ). On a leakage-free holdout of 47 official episodes, mixing official data into Stage 2 (Mix-FT) matches the synthetic-only fine-tune on FR (0.713 vs. 0.709) while cutting NQ from 3.45 to 0.67. The submitted weights (Full-FT) additionally absorb the holdout episodes with the same recipe. Ablations cover oracle identity, prompt family, backbone choice (including a full Qwen3.6-27B replica that does *not* transfer), ask/conclude ratio, and a rejected chain-rebalancing branch.

1 Task, Metrics, and Protocol

Given a natural-language description of a target object and a stream of candidate images, the agent must accept the candidate as the target instance (score 2), reject it (score 0), or ask the oracle one clarifying question (score 1) [2, 1]. Official training data comprise 167 episodes (50 categories, 367 images) and six description types: category, color, context, color+feature, color+context, and color+context+feature.

Metrics. SR is the per-episode fraction of correct decisions, averaged over episodes: the denominator is the number of successes if the episode is fully solved, otherwise successes+1 (the episode stops at the first error). FR is the fraction of episodes in which every decision is correct. NQ/obs is the per-episode ratio of questions to observations, then averaged. Ask rate is the fraction of episodes with at least one question. We do *not* use a global $\sum$successes/$\sum$observations ratio: that statistic weights long episodes and can exceed 1 when observations are de-duplicated by pixel identity.

Selection rule. FR first, SR second, lower NQ as tie-break. All reported numbers use temperature 0. Unless noted, development uses a frozen local oracle, Qwen3-VL-32B-Instruct. Model selection never trains on a 47-episode holdout obtained by splitting official episodes along shared-image connected components (zero image leakage).

Naming. For readability we use paper names rather than checkpoint identifiers. The frozen inference/data prompt is the *Structured Attribute Prompt* (SAP). Fine-tuned 32B variants are Syn-FT (synthetic Stage 2 only), Mix-FT (Stage 2 continued on official train-split pairs), Chain-FT (a rejected rebalancing branch), and Full-FT (submitted; Mix-FT recipe with holdout pairs included). Release adapters in the package are labeled `v3d/v3o/v3p/v3f` respectively.

2 Oracle Identity

With the challenge-provided 7B questioner and the paper prompt under raw decoding, we compared Qwen2.5-VL-32B-Instruct and Qwen3-VL-32B-Instruct as oracles on all 167 episodes (Table 1). Mean $|\Delta\mathrm{SR}|$ is below 0.01 on every type. Under this low-ask regime the questioner almost never queries the oracle, so oracle identity cannot move the metric. Qwen3 is slightly better on color/context and slightly worse on category; we freeze Qwen3-VL-32B as the development oracle thereafter. The practical implication is that oracle quality matters mainly in high-ask policies and in *labeling* (Section 5), not in a silent baseline.

Type	Qwen2.5-32B SR/FR/NQ	Qwen3-32B SR/FR/NQ	ΔSR
category	0.280/0.263/0.009	0.274/0.257/0.009	−0.006
color	0.591/0.485/0.058	0.594/0.497/0.068	+0.003
context	0.567/0.439/0.156	0.574/0.445/0.146	+0.007
color+feature	0.712/0.571/0.138	0.707/0.574/0.155	−0.005
color+context	0.736/0.634/0.201	0.741/0.638/0.198	+0.004
color+ctx+feat.	0.712/0.586/0.267	0.706/0.589/0.249	−0.005

Table 1: Oracle swap with a low-ask paper prompt and the 7B challenge questioner, full 167 episodes. Differences are negligible.

3 Prompt Selection

All prompts share a three-field output schema: a `<motivation>` span, a `<score>` in $\{0, 1, 2\}$, and a `<question>` that is either a single query or `None`. We iterate the *instruction* while holding the schema fixed, so that supervised labels and inference stay aligned.

3.1 Paper family on the 7B questioner

Table 2 averages six types (full 167 episodes, Qwen3 oracle, raw decoding). *Paper* is the QAsk-Nav wording. *Paper-YN* adds a yes/no discriminative-question constraint. *Adaptive* writes an information-dependent ask/conclude rule into a single prompt (ask when the description is underspecified; conclude when attributes already suffice). *Uncertainty gating* forces an ask when the model emits a confident score but hesitant wording.

Prompt / policy	SR	FR	NQ/obs
Paper + raw decoding	0.600	0.500	0.138
Paper-YN + raw decoding	0.616	0.491	0.088
Adaptive + raw decoding	0.615	0.491	0.089
Paper + uncertainty gating	0.719	0.562	0.838

Table 2: Prompt and policy means on the 7B questioner (six types, 167 episodes). Uncertainty gating is discarded as a spurious gain.

Uncertainty gating looks best on SR/FR but inflates NQ six-fold. About 70% of its questions are a canned template ("Besides what is already stated in …"). The SR lift comes from forcing a second look at the image, not from informative questions. We discard it.

Paper-YN and Adaptive are essentially tied and both beat Paper. Adaptive matches Paper-YN without a per-type switch: category/context SR rise (+0.029 / +0.040 vs. Paper) while NQ on attribute-rich types falls (e.g. color 0.068 → 0.042). Questions under Adaptive on color+feature are genuine yes/no checks, not templates. We therefore take Adaptive as the best *single-prompt* member of the paper family.

3.2 Why a new prompt is needed

Two failures of Adaptive motivate a rewrite. First, "the description is already specific, so do not ask" treats text informativeness as a sufficient statistic for the decision. A specific attribute that is occluded in the current view still requires a question. Second, discriminative questions anchored on the *candidate* image do not transfer to the next view; the official protocol instead asks the oracle about the hidden *target* image in order to build a reusable instance profile [2].

3.3 Profile prompts, then SAP

Profile-V1 forces OBSERVE → COMPARE → DECIDE in `<motivation>`, scores each stated attribute as MATCHED / CONTRADICTED / UNVERIFIABLE, and requires questions to request a new target-side attribute that is also checkable on candidates. Putting Profile-V1 on the 7B fine-tune *without* relabeling fails: the model ignores the three-step format, over-asks on specific types, and on long contexts degenerates into repeated tokens. The same prompt on *native* Qwen3-VL-32B succeeds (Table 3): category SR 0.340 → 0.436, color+feature 0.726 → 0.753, with explicit Step 1/2/3 throughout and no loops. The 7B failure was train/test prompt mismatch, not a bad prompt.

Profile-V2 compresses reasoning ($<$ 120 words) and forbids score 2 unless a distinguishing attribute is present. On native 32B it is uniformly worse than V1 (mean SR 0.694 vs. 0.709, FR 0.608 vs. 0.633) except for a small category SR bump paid for with much higher NQ. Compactness alone is not an improvement.

An intermediate 32B fine-tune on Profile-V2 labels raised category SR from native V1's 0.436 to 0.529 but lagged on several specific types. That split—gains concentrated on type-only descriptions, losses on richer ones—motivated rebuilding labels under a stricter protocol rather than shipping the V2 model.

SAP (frozen for all later data and inference) keeps V1's three-step compare protocol and V2's length cap, and adds stricter decision rules: score 0 if the described type is absent or any stated attribute is contradicted; score 1 if the description is type-only or any stated attribute is unverifiable; score 2 only when every *stated* attribute matches and none is unverifiable. Questions ask exactly one new target attribute. On native 32B, SAP raises mean SR/FR to 0.739/0.653 with NQ 0.402 (category NQ 1.11, ask rate 99%). It is not uniformly best versus V1 on every attribute-rich type (V1 files there are slightly incomplete, $n = 127$–164), but it is the protocol we need in *labels*: every stated attribute is audited, type-only descriptions must ask, and the same text is used at train and test. We freeze SAP from data generation onward.

Setting (native 32B, raw decoding)	SR	FR	NQ/obs	Ask rate
Adaptive, category	0.340	0.311	0.162	22%
Profile-V1, category	**0.436**	**0.395**	0.436	49%
Adaptive, color+feature	0.726	0.640	0.030	7%
Profile-V1, color+feature	**0.753**	**0.659**	0.077	9%
Profile-V1 (six-type mean[†])	**0.709**	**0.633**	0.128	16%
Profile-V2 (six types, n=167)	0.694	0.608	0.164	20%
SAP (six types, n=167)	0.739	0.653	0.402	40%

Table 3: Native Qwen3-VL-32B prompt comparison. [†]Some V1 attribute-type files are slightly incomplete ($n = 127$–164); the mean is the best available V1 estimate.

4 Backbone Selection

Replacing the 7B challenge questioner with native Qwen3-VL-32B under SAP is the first large jump (Section 3). We then swap only the questioner to native Qwen3.6-27B, keeping SAP, raw decoding, temperature 0, and the same 32B oracle (Table 4). Mean SR/FR rise (0.739 → 0.748, 0.653 → 0.665) and NQ drops 55% (0.402 → 0.182). Five of six types improve; only context regresses. Category NQ falls from 1.11 to 0.70. (Category n=166 for Qwen3.6, one missing stove episode; other types n=167.) Native Qwen3.6 is the stronger *zero-shot* questioner.

We nevertheless fine-tune Qwen3-VL-32B: the QLoRA, image-size, and serving stack were validated on that architecture, and a stronger zero-shot prior does not imply that an ask-heavy supervised recipe will transfer. Section 7.3 reports a full replica that loses on holdout FR; the submitted backbone remains 32B.

Type	Qwen3.6-27B SR/FR/NQ	Qwen3-VL-32B SR/FR/NQ
category	0.679/0.590/0.699	0.659/0.575/1.114
color	0.682/0.599/0.066	0.672/0.557/0.295
context	0.648/0.557/0.146	0.685/0.605/0.351
color+feature	0.768/0.683/0.046	0.742/0.653/0.351
color+context	0.845/0.767/0.068	0.836/0.749/0.172
color+ctx+feat.	0.868/0.796/0.071	0.839/0.778/0.127
Mean	**0.748/0.665/0.182**	0.739/0.653/0.402

Table 4: Zero-shot SAP, same oracle and policy, full official set. Qwen3.6 asks less and scores slightly higher.

5 Data Construction

Labels are generated with SAP frozen, so the supervised distribution matches inference. We never train on raw code-variable dumps; each sample is a (description, candidate image, optional dialogue, motivation, score, question) tuple.

5.1 Synthetic instances

Candidate images come primarily from Habitat renders of CoIN-Bench / HM3D episodes (target views and nearby viewpoints), plus same-category cross-instance pairing as hard negatives. InstructPix2Pix appearance edits were prototyped and then demoted: geometric renders are the main source. Each pair is labeled by Gemini Flash under SAP. Strata with low Flash/Pro agreement are escalated to Gemini Pro (Pro is the final arbiter); invalid or persistently conflicting items are quarantined rather than force-labeled.

Multi-turn chains (0–6 rounds) are collected with Flash as questioner and Pro as answerer against the target reference. We then delete degenerate behavior: near-duplicate questions, answers that contradict already-stated history, and reasoning that ignores available context (the dominant failure mode of an early fine-tune). The train/validation cut is at instance level, not image crop level.

5.2 Two-stage supervision

Following QAsk-Nav [2]: Stage 1 trains motivation plus score on all labeled pairs (20,653 examples) with empty question slots, to stabilize the schema. Stage 2 continues on the full schema with an ask-weighted mix. A 10:1 ask:conclude branch is dominated by 5:1 on a 60-episode slice (SR 0.773 / FR 0.669 / NQ 7.12 vs. 0.806 / 0.700 / 4.63); extra ask mass amplifies repeat questions. We keep 5:1 (8,259 Stage 2 examples). Training is QLoRA, 4-bit NF4, LoRA rank 16 ($\approx$ 40M trainable parameters on 32B), images resized to 224^2, 7-GPU DDP, learning rate 1×10^{-4} for Stages 1–2 and 5×10^{-5} for official continuations, one epoch per stage.

5.3 Official episodes, without leakage

Synthetic images do not overlap the official 367 files. To use official episodes without contaminating the test of generalization, we split the 167 episodes by shared-image connected components into 120 train / 47 holdout episodes. Holdout images never appear in Mix-FT training.

On the 120-episode split we label all description×candidate pairs (2,298) with Flash under SAP and *validate against episode ground truth*:

- agreeing conclusions are kept (1,484);
- Flash-match vs. GT-reject conflicts (296 of 302 disagreements) are dropped from conclusion training—these are exactly the underspecified cases the benchmark wants the agent to *ask* about, so teaching a hard accept would be label noise;
- depth-0 ask outputs are kept (496).

This yields 1,980 in-domain examples mixed into Stage 2 (10,239 total) for a one-epoch continuation from Syn-FT (Mix-FT).

After selecting Mix-FT we apply the same pipeline to the 47 holdout episodes (870 pairs → 716 kept: 190 ask / 229 match / 297 reject) and continue from the same Syn-FT initialization (Full-FT; 10,955 Stage 2 examples). Selection numbers always cite Mix-FT on holdout; Full-FT is the submitted weight.

5.4 A rejected rebalancing branch

Mix-FT regresses on context and color+feature, consistent with official conclusions suppressing questions. We built 636 GT-checked one-round chains (138 forced asks, 355 contextual rejects, 143 matches) and down-sampled official rejects to 1.2× matches (Chain-FT). Holdout FR fell to 0.677 (Table 5). At this scale the natural official label mix is closer to optimal than hand-rebalanced chains. Chain-FT is not used.

6 Inference Policy

Syn-FT on category-only descriptions enters repeat-ask loops. A lightweight rule detects near-duplicate questions (Jaccard ≥ 0.6 over content tokens of length ≥ 2) or budget exhaustion, then re-prompts once to decide from answers already collected (fallback: reject). Applying this rule to *all* types cuts NQ to 0.74 but costs up to 13pp FR on color. Restricting it to category is strictly helpful on the full 167 set (category SR 0.663 → 0.725, FR 0.569 → 0.641, NQ 10.2 → 1.05). All reported fine-tune numbers use this hybrid policy: forced decision on category, raw decoding otherwise.

7 Main Results and Selection

7.1 Holdout-47 (clean benchmark)

Table 5 is the selection table. Syn-FT raises FR over zero-shot SAP but explodes NQ. Mix-FT keeps FR (0.713 vs. 0.709) and reduces NQ five-fold. Per type, official data helps category (FR 0.596 → 0.745, SR 0.713 → 0.840) and color+context (FR 0.766 → 0.830), with some loss on context and color+feature. Under the pre-registered rule Mix-FT wins. Chain-FT is dominated on FR.

Qwen3-VL-32B, hybrid policy, holdout-47	SR	FR	NQ/obs
Zero-shot + SAP	0.766	0.688	0.41
Syn-FT	**0.823**	0.709	3.45
Mix-FT (selected)	0.801	**0.713**	0.67
Chain-FT (not used)	0.772	0.677	0.54

Table 5: Leakage-free holdout, six types averaged. Mix-FT is selected by FR, then SR, then NQ.

Type	Syn-FT SR/FR/NQ	Mix-FT SR/FR/NQ
category	0.713/0.596/0.98	**0.840**/**0.745**/0.83
color	0.836/0.681/5.58	0.779/0.681/**1.20**
context	**0.881**/**0.787**/3.21	0.720/0.660/0.56
color+feature	0.817/0.660/3.11	0.755/0.596/0.53
color+context	0.857/0.766/3.72	0.872/**0.830**/0.70
color+ctx+feat.	0.833/0.766/4.09	0.840/0.766/**0.18**
Mean	0.823/0.709/3.45	0.801/**0.713**/**0.67**

Table 6: Holdout-47 by description type. Official pairs mainly repair category behavior and suppress redundant questions.

7.2 Full-FT sanity

Continuing from Syn-FT with all 167 official episodes included, a same-slice sanity on the first 60 episodes per type shows no regression vs. Mix-FT: Mix-FT SR/FR/NQ 0.768/0.667/0.68 vs. Full-FT 0.768/0.678/0.61, with category FR 0.517 → 0.633. Full-FT is the submitted weight. Clean-benchmark numbers remain Mix-FT on holdout-47.

On the *full* 167 episodes, Syn-FT plus the hybrid policy already beats native SAP (mean SR 0.814 / FR 0.714 / NQ 3.88 vs. 0.739/0.653/0.40). Mix-FT/Full-FT exist to keep that accuracy while making NQ competition-plausible.

7.3 Replica on Qwen3.6-27B

We repeat Stage 1, Stage 2 (5:1), and Mix-FT on Qwen3.6-27B with the same files, hyperparameters, holdout, SAP, and hybrid policy (Table 7). Native Qwen3.6 was stronger; after the same SFT recipe both Syn-FT and Mix-FT trail the 32B counterparts on FR and SR. Mix-FT NQ is competitive (0.54 vs. 0.67) but cannot overturn the FR gap (−6.4pp). We therefore do not switch the submission. The likely cause is a mismatch between an already-conservative zero-shot prior and ask-heavy Stage 2 data, not a failed load: a smoke test of the 3.6 adapter produces well-formed SAP outputs.

Holdout-47, hybrid policy	Qwen3.6-27B	Qwen3-VL-32B
Syn-FT mean SR/FR/NQ	0.779/0.642/3.15	**0.823**/**0.709**/3.45
Mix-FT mean SR/FR/NQ	0.768/0.649/0.54	**0.801**/**0.713**/0.67

Table 7: Same data and recipe on two backbones. Fine-tuning does not preserve Qwen3.6's zero-shot advantage; 32B Mix-FT remains selected.

8 Submitted System

Questioner: Qwen3-VL-32B-Instruct with the Full-FT LoRA adapter (rank 16), SAP, temperature 0, hybrid policy (forced decision only on category-only descriptions). Development oracle: Qwen3-VL-32B-Instruct. The package includes the adapter, optional merged bf16 weights, Mix-FT as a clean-benchmark backup, data and training scripts, and the frozen prompt. Serve with vLLM `--enable-lora` on the bf16 base, or serve the merged checkpoint directly.

9 Reproducibility Notes

Connected-component episode IDs, Stage 2 manifests, and the exact SAP string live in the code directory. Metrics are recomputed from dumped JSON with a single script that implements Section 1. Continuation runs always start from Syn-FT, not from Mix-FT, so Mix-FT and Full-FT are parallel forks rather than a chain.

References

[1] F. Taioli, E. Zorzi, G. Franchi, A. Castellini, A. Farinelli, M. Cristani, Y. Wang. Collaborative Instance Object Navigation: Leveraging Uncertainty-Awareness to Minimize Human-Agent Dialogues. In *ICCV*, 2025.

[2] E. Zorzi, F. Taioli, Y. Wang, M. Cristani, A. Farinelli, A. Castellini, L. Bazzani. Benchmarking Interaction, Beyond Policy: a Reproducible Benchmark for Collaborative Instance Object Navigation. arXiv:2604.00265, 2026.